PG No. 27 D_{6h} $6/mmm$ [hexagonal] (internal axial octupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(3,3)}[g](A_{1u})$

** symmetry

1

** expression

$$\frac{\sqrt{70}G_{3c}x\left(x^2-3y^2\right)}{28}+\frac{\sqrt{70}G_{3s}y\left(3x^2-y^2\right)}{28}-\frac{\sqrt{42}G_{3x}x\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{42}G_{3y}y\left(x^2+y^2-4z^2\right)}{28}$$
$$+\frac{\sqrt{105}G_{xyz}xyz}{7}-\frac{\sqrt{7}G_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{105}G_z^{\beta}z\left(x-y\right)\left(x+y\right)}{14}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(3,1)}[g](A_{2g})$

** symmetry

z

** expression

$$\frac{2\sqrt{210}G_{3x}xz}{35}+\frac{2\sqrt{210}G_{3y}yz}{35}+\frac{\sqrt{21}G_{xyz}xy}{7}-\frac{3\sqrt{35}G_z^{\alpha}\left(x^2+y^2-2z^2\right)}{70}+\frac{\sqrt{21}G_z^{\beta}\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](A_{2g})$

** symmetry

z

** expression

$$\frac{\sqrt{210}G_{3c}xz\left(x^2-3y^2\right)}{24}+\frac{\sqrt{210}G_{3s}yz\left(3x^2-y^2\right)}{24}-\frac{5\sqrt{14}G_{3x}xz\left(3x^2+3y^2-4z^2\right)}{56}-\frac{5\sqrt{14}G_{3y}yz\left(3x^2+3y^2-4z^2\right)}{56}$$
$$-\frac{\sqrt{35}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{14}+\frac{\sqrt{21}G_z^{\alpha}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{84}-\frac{\sqrt{35}G_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{28}$$

$\vec{\mathbb{G}}_{1,1}^{(3,1)}[g](E_{1g})$, $\vec{\mathbb{G}}_{1,2}^{(3,1)}[g](E_{1g})$

** symmetry

x

y

** expression

$$\frac{3\sqrt{14}G_{3c}\left(x-y\right)\left(x+y\right)}{28}+\frac{3\sqrt{14}G_{3s}xy}{14}-\frac{\sqrt{210}G_{3x}\left(3x^2+y^2-4z^2\right)}{140}-\frac{\sqrt{210}G_{3y}xy}{70}+\frac{\sqrt{21}G_{xyz}yz}{7}-\frac{3\sqrt{35}G_z^{\alpha}xz}{35}+\frac{\sqrt{21}G_z^{\beta}xz}{7}$$

$$-\frac{3\sqrt{14}G_{3c}xy}{14}+\frac{3\sqrt{14}G_{3s}\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{210}G_{3x}xy}{70}-\frac{\sqrt{210}G_{3y}\left(x^2+3y^2-4z^2\right)}{140}+\frac{\sqrt{21}G_{xyz}xz}{7}-\frac{3\sqrt{35}G_z^{\alpha}yz}{35}-\frac{\sqrt{21}G_z^{\beta}yz}{7}$$

$\vec{\mathbb{G}}_{1,1}^{(3,3)}[g](E_{1g})$, $\vec{\mathbb{G}}_{1,2}^{(3,3)}[g](E_{1g})$

** symmetry

x

y

** expression

$$\frac{\sqrt{210}G_{3c}\left(4x^4-21x^2y^2-3x^2z^2+3y^4+3y^2z^2\right)}{168}+\frac{\sqrt{210}G_{3s}xy\left(15x^2-13y^2-6z^2\right)}{168}-\frac{\sqrt{14}G_{3x}\left(4x^4+3x^2y^2-27x^2z^2-y^4+3y^2z^2+4z^4\right)}{56}$$
$$-\frac{5\sqrt{14}G_{3y}xy\left(x^2+y^2-6z^2\right)}{56}+\frac{\sqrt{35}G_{xyz}yz\left(6x^2-y^2-z^2\right)}{14}-\frac{5\sqrt{21}G_z^{\alpha}xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_z^{\beta}xz\left(5x^2-9y^2-2z^2\right)}{28}$$

$$\frac{\sqrt{210}G_{3c}xy\left(13x^2-15y^2+6z^2\right)}{168}-\frac{\sqrt{210}G_{3s}\left(3x^4-21x^2y^2+3x^2z^2+4y^4-3y^2z^2\right)}{168}$$
$$-\frac{5\sqrt{14}G_{3x}xy\left(x^2+y^2-6z^2\right)}{56}+\frac{\sqrt{14}G_{3y}\left(x^4-3x^2y^2-3x^2z^2-4y^4+27y^2z^2-4z^4\right)}{56}$$
$$-\frac{\sqrt{35}G_{xyz}xz\left(x^2-6y^2+z^2\right)}{14}-\frac{5\sqrt{21}G_z^{\alpha}yz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_z^{\beta}yz\left(9x^2-5y^2+2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](A_{2u})$

** symmetry

z

** expression

$$\frac{3\sqrt{70}G_{3c}y\left(3x^2-y^2\right)}{56}-\frac{3\sqrt{70}G_{3s}x\left(x^2-3y^2\right)}{56}-\frac{\sqrt{42}G_{3x}y\left(x^2+y^2-4z^2\right)}{56}+\frac{\sqrt{42}G_{3y}x\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{105}G_{xyz}z\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_{z}^{\beta}xyz}{7}$$

$\vec{\mathbb{Q}}_{1,1}^{(3,2)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{1,2}^{(3,2)}[g](E_{1u})$
** symmetry

x

y

** expression

$$-\frac{3\sqrt{70}G_{3c}xyz}{28}+\frac{3\sqrt{70}G_{3s}z\left(x-y\right)\left(x+y\right)}{56}-\frac{5\sqrt{42}G_{3x}xyz}{28}-\frac{\sqrt{42}G_{3y}z\left(x^2+11y^2-4z^2\right)}{56}$$
$$-\frac{\sqrt{105}G_{xyz}x\left(y-z\right)\left(y+z\right)}{14}+\frac{3\sqrt{7}G_{z}^{\alpha}y\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{105}G_{z}^{\beta}y\left(x^2-y^2+2z^2\right)}{28}$$

$$-\frac{3\sqrt{70}G_{3c}z\left(x-y\right)\left(x+y\right)}{56}-\frac{3\sqrt{70}G_{3s}xyz}{28}+\frac{\sqrt{42}G_{3x}z\left(11x^2+y^2-4z^2\right)}{56}+\frac{5\sqrt{42}G_{3y}xyz}{28}$$
$$+\frac{\sqrt{105}G_{xyz}y\left(x-z\right)\left(x+z\right)}{14}-\frac{3\sqrt{7}G_{z}^{\alpha}x\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}G_{z}^{\beta}x\left(x^2-y^2-2z^2\right)}{28}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_{2}^{(3,0)}[g](A_{1g})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{21}G_{3x}yz}{7}-\frac{\sqrt{21}G_{3y}xz}{7}-\frac{\sqrt{210}G_{xyz}\left(x-y\right)\left(x+y\right)}{28}+\frac{\sqrt{210}G_{z}^{\beta}xy}{14}$$

$\vec{\mathbb{Q}}_{2}^{(3,2)}[g](A_{1g})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{70}G_{3c}yz\left(3x^2-y^2\right)}{8}-\frac{\sqrt{70}G_{3s}xz\left(x^2-3y^2\right)}{8}-\frac{5\sqrt{42}G_{3x}yz\left(3x^2+3y^2-4z^2\right)}{168}$$
$$+\frac{5\sqrt{42}G_{3y}xz\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{105}G_{xyz}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{42}-\frac{\sqrt{105}G_{z}^{\beta}xy\left(x^2+y^2-6z^2\right)}{21}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,0)}[g](E_{1g})$, $\vec{\mathbb{Q}}_{2,2}^{(3,0)}[g](E_{1g})$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$\frac{\sqrt{105}G_{3c}\left(x-y\right)\left(x+y\right)}{28}+\frac{\sqrt{105}G_{3s}xy}{14}+\frac{\sqrt{7}G_{3x}\left(5x^2-y^2-4z^2\right)}{28}+\frac{3\sqrt{7}G_{3y}xy}{14}+\frac{\sqrt{42}G_{z}^{\alpha}xz}{7}$$

$$-\frac{\sqrt{105}G_{3c}xy}{14}+\frac{\sqrt{105}G_{3s}\left(x-y\right)\left(x+y\right)}{28}+\frac{3\sqrt{7}G_{3x}xy}{14}-\frac{\sqrt{7}G_{3y}\left(x^2-5y^2+4z^2\right)}{28}+\frac{\sqrt{42}G_{z}^{\alpha}yz}{7}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,2)}[g](E_{1g})$, $\vec{\mathbb{Q}}_{2,2}^{(3,2)}[g](E_{1g})$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$-\frac{\sqrt{210}G_{3c}\left(2x^4-21x^2y^2+9x^2z^2+5y^4-9y^2z^2\right)}{168}-\frac{\sqrt{210}G_{3s}xy\left(11x^2-17y^2+18z^2\right)}{168}-\frac{5\sqrt{14}G_{3x}\left(2x^4+3x^2y^2-15x^2z^2+y^4-9y^2z^2+4z^4\right)}{168}$$
$$-\frac{5\sqrt{14}G_{3y}xy\left(x^2+y^2-6z^2\right)}{168}+\frac{\sqrt{35}G_{xyz}yz\left(y-z\right)\left(y+z\right)}{6}-\frac{5\sqrt{21}G_{z}^{\alpha}xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_{z}^{\beta}xz\left(x^2+3y^2-2z^2\right)}{12}$$

$$-\frac{\sqrt{210}G_{3c}xy\left(17x^2-11y^2-18z^2\right)}{168}+\frac{\sqrt{210}G_{3s}\left(5x^4-21x^2y^2-9x^2z^2+2y^4+9y^2z^2\right)}{168}-\frac{5\sqrt{14}G_{3x}xy\left(x^2+y^2-6z^2\right)}{168}$$
$$-\frac{5\sqrt{14}G_{3y}\left(x^4+3x^2y^2-9x^2z^2+2y^4-15y^2z^2+4z^4\right)}{168}+\frac{\sqrt{35}G_{xyz}xz\left(x-z\right)\left(x+z\right)}{6}-\frac{5\sqrt{21}G_z^{\alpha}yz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{\sqrt{35}G_z^{\beta}yz\left(3x^2+y^2-2z^2\right)}{12}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,0)}[g](E_{2g})$, $\vec{\mathbb{Q}}_{2,2}^{(3,0)}[g](E_{2g})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{105}G_{3c}yz}{14}+\frac{\sqrt{105}G_{3s}xz}{14}-\frac{3\sqrt{7}G_{3x}yz}{14}-\frac{3\sqrt{7}G_{3y}xz}{14}-\frac{\sqrt{70}G_{xyz}\left(x^2+y^2-2z^2\right)}{28}+\frac{\sqrt{42}G_z^{\alpha}xy}{14}$$

$$\frac{\sqrt{105}G_{3c}xz}{14}+\frac{\sqrt{105}G_{3s}yz}{14}-\frac{3\sqrt{7}G_{3x}xz}{14}+\frac{3\sqrt{7}G_{3y}yz}{14}+\frac{\sqrt{42}G_z^{\alpha}\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{70}G_z^{\beta}\left(x^2+y^2-2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,2)}[g](E_{2g})$, $\vec{\mathbb{Q}}_{2,2}^{(3,2)}[g](E_{2g})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{210}G_{3c}yz\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{210}G_{3s}xz\left(3x^2+3y^2-4z^2\right)}{168}-\frac{5\sqrt{14}G_{3x}yz\left(27x^2-y^2-8z^2\right)}{168}+\frac{5\sqrt{14}G_{3y}xz\left(x^2-27y^2+8z^2\right)}{168}$$
$$+\frac{\sqrt{35}G_{xyz}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{42}+\frac{5\sqrt{21}G_z^{\alpha}xy\left(x^2+y^2-6z^2\right)}{42}-\frac{\sqrt{35}G_z^{\beta}xy\left(x-y\right)\left(x+y\right)}{6}$$

$$\frac{\sqrt{210}G_{3c}xz\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{210}G_{3s}yz\left(3x^2+3y^2-4z^2\right)}{168}-\frac{5\sqrt{14}G_{3x}xz\left(13x^2-15y^2-8z^2\right)}{168}-\frac{5\sqrt{14}G_{3y}yz\left(15x^2-13y^2+8z^2\right)}{168}$$
$$-\frac{\sqrt{35}G_{xyz}xy\left(x-y\right)\left(x+y\right)}{6}+\frac{5\sqrt{21}G_z^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{84}-\frac{\sqrt{35}G_z^{\beta}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{84}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{14}G_{3x}x}{7}+\frac{\sqrt{14}G_{3y}y}{7}+\frac{\sqrt{21}G_z^{\alpha}z}{7}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{5\sqrt{210}G_{3c}x\left(x^2-3y^2\right)}{168}-\frac{5\sqrt{210}G_{3s}y\left(3x^2-y^2\right)}{168}-\frac{3\sqrt{14}G_{3x}x\left(x^2+y^2-4z^2\right)}{56}-\frac{3\sqrt{14}G_{3y}y\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{21}G_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{1155}G_{3c}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{264}-\frac{\sqrt{1155}G_{3s}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{264}+\frac{15\sqrt{77}G_{3x}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{616}$$
$$+\frac{15\sqrt{77}G_{3y}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{616}-\frac{3\sqrt{770}G_{xyz}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{5\sqrt{462}G_z^{\alpha}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1848}-\frac{3\sqrt{770}G_z^{\beta}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$

$\vec{\mathbb{G}}_{2,1}^{(3,-1)}[g](E_{1u})$, $\vec{\mathbb{G}}_{2,2}^{(3,-1)}[g](E_{1u})$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{2\sqrt{42}G_{3y}z}{21}+\frac{\sqrt{105}G_{xyz}x}{21}-\frac{\sqrt{7}G_z^{\alpha}y}{7}-\frac{\sqrt{105}G_z^{\beta}y}{21}$$

$$-\frac{2\sqrt{42}G_{3x}z}{21}-\frac{\sqrt{105}G_{xyz}y}{21}+\frac{\sqrt{7}G_z^{\alpha}x}{7}-\frac{\sqrt{105}G_z^{\beta}x}{21}$$

$\vec{\mathbb{G}}_{2,1}^{(3,1)}[g](E_{1u})$, $\vec{\mathbb{G}}_{2,2}^{(3,1)}[g](E_{1u})$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$-\frac{5\sqrt{70}G_{3c}xyz}{28}+\frac{5\sqrt{70}G_{3s}z\left(x-y\right)\left(x+y\right)}{56}+\frac{5\sqrt{42}G_{3x}xyz}{28}-\frac{\sqrt{42}G_{3y}z\left(21x^2-9y^2-4z^2\right)}{168}$$
$$-\frac{\sqrt{105}G_{xyz}x\left(2x^2-3y^2-3z^2\right)}{42}-\frac{\sqrt{7}G_z^{\alpha}y\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}G_z^{\beta}y\left(9x^2-y^2-6z^2\right)}{84}$$

$$-\frac{5\sqrt{70}G_{3c}z\left(x-y\right)\left(x+y\right)}{56}-\frac{5\sqrt{70}G_{3s}xyz}{28}-\frac{\sqrt{42}G_{3x}z\left(9x^2-21y^2+4z^2\right)}{168}-\frac{5\sqrt{42}G_{3y}xyz}{28}$$
$$-\frac{\sqrt{105}G_{xyz}y\left(3x^2-2y^2+3z^2\right)}{42}+\frac{\sqrt{7}G_z^{\alpha}x\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{105}G_z^{\beta}x\left(x^2-9y^2+6z^2\right)}{84}$$

$\vec{\mathbb{G}}_{2,1}^{(3,3)}[g](E_{1u})$, $\vec{\mathbb{G}}_{2,2}^{(3,3)}[g](E_{1u})$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{385}G_{3c}xyz\left(5x^2-7y^2+2z^2\right)}{44}-\frac{\sqrt{385}G_{3s}z\left(x^4-9x^2y^2+x^2z^2+2y^4-y^2z^2\right)}{44}-\frac{5\sqrt{231}G_{3x}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{5\sqrt{231}G_{3y}z\left(3x^4-15x^2y^2-x^2z^2-18y^4+41y^2z^2-4z^4\right)}{924}+\frac{\sqrt{2310}G_{xyz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{462}$$
$$+\frac{5\sqrt{154}G_z^{\alpha}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}-\frac{\sqrt{2310}G_z^{\beta}y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{924}$$

$$-\frac{\sqrt{385}G_{3c}z\left(2x^4-9x^2y^2-x^2z^2+y^4+y^2z^2\right)}{44}-\frac{\sqrt{385}G_{3s}xyz\left(7x^2-5y^2-2z^2\right)}{44}+\frac{5\sqrt{231}G_{3x}z\left(18x^4+15x^2y^2-41x^2z^2-3y^4+y^2z^2+4z^4\right)}{924}$$
$$+\frac{5\sqrt{231}G_{3y}xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{\sqrt{2310}G_{xyz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{462}$$
$$-\frac{5\sqrt{154}G_z^{\alpha}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}+\frac{\sqrt{2310}G_z^{\beta}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{924}$$

$\vec{\mathbb{G}}_{2,1}^{(3,-1)}[g](E_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(3,-1)}[g](E_{2u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$\frac{\sqrt{70}G_{3c}x}{14}+\frac{\sqrt{70}G_{3s}y}{14}-\frac{\sqrt{42}G_{3x}x}{42}+\frac{\sqrt{42}G_{3y}y}{42}+\frac{\sqrt{105}G_z^{\beta}z}{21}$$

$$\frac{\sqrt{70}G_{3c}y}{14}-\frac{\sqrt{70}G_{3s}x}{14}+\frac{\sqrt{42}G_{3x}y}{42}+\frac{\sqrt{42}G_{3y}x}{42}-\frac{\sqrt{105}G_{xyz}z}{21}$$

$\vec{\mathbb{G}}_{2,1}^{(3,1)}[g](E_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(3,1)}[g](E_{2u})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$\frac{\sqrt{70}G_{3c}x\left(x^2+y^2-4z^2\right)}{56} + \frac{\sqrt{70}G_{3s}y\left(x^2+y^2-4z^2\right)}{56} - \frac{\sqrt{42}G_{3x}x\left(11x^2-9y^2-24z^2\right)}{168}$$
$$- \frac{\sqrt{42}G_{3y}y\left(9x^2-11y^2+24z^2\right)}{168} - \frac{5\sqrt{7}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)}{14} + \frac{\sqrt{105}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{42}$$

$$\frac{\sqrt{70}G_{3c}y\left(x^2+y^2-4z^2\right)}{56} - \frac{\sqrt{70}G_{3s}x\left(x^2+y^2-4z^2\right)}{56} + \frac{\sqrt{42}G_{3x}y\left(21x^2+y^2-24z^2\right)}{168}$$
$$+ \frac{\sqrt{42}G_{3y}x\left(x^2+21y^2-24z^2\right)}{168} - \frac{\sqrt{105}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{42} + \frac{5\sqrt{7}G_z^{\alpha}xyz}{7}$$

$\vec{\mathbb{G}}_{2,1}^{(3,3)}[g](E_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(3,3)}[g](E_{2u})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$\frac{\sqrt{385}G_{3c}x\left(11x^4-104x^2y^2-6x^2z^2+53y^4-6y^2z^2+4z^4\right)}{616} + \frac{\sqrt{385}G_{3s}y\left(53x^4-104x^2y^2-6x^2z^2+11y^4-6y^2z^2+4z^4\right)}{616}$$
$$- \frac{5\sqrt{231}G_{3x}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{1848} - \frac{5\sqrt{231}G_{3y}y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{1848}$$
$$+ \frac{3\sqrt{2310}G_{xyz}xyz\left(x-y\right)\left(x+y\right)}{44} - \frac{5\sqrt{154}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88} + \frac{\sqrt{2310}G_z^{\beta}z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{1848}$$

$$- \frac{\sqrt{385}G_{3c}y\left(26x^4-53x^2y^2+3x^2z^2+5y^4+3y^2z^2-2z^4\right)}{308} + \frac{\sqrt{385}G_{3s}x\left(5x^4-53x^2y^2+3x^2z^2+26y^4+3y^2z^2-2z^4\right)}{308}$$
$$+ \frac{5\sqrt{231}G_{3x}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{924} - \frac{5\sqrt{231}G_{3y}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{924}$$
$$+ \frac{\sqrt{2310}G_{xyz}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{462} + \frac{5\sqrt{154}G_z^{\alpha}xyz\left(x^2+y^2-2z^2\right)}{44} - \frac{3\sqrt{2310}G_z^{\beta}xyz\left(x-y\right)\left(x+y\right)}{44}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_{3}^{(3,-3)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$G_z^{\alpha}$$

$\vec{\mathbb{G}}_{3}^{(3,-1)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{10}G_{3x}xz}{10} + \frac{\sqrt{10}G_{3y}yz}{10} - G_{xyz}xy - \frac{\sqrt{15}G_z^{\alpha}\left(x^2+y^2-2z^2\right)}{15} - \frac{G_z^{\beta}\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_{3}^{(3,1)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{7\sqrt{55}G_{3c}xz\left(x^2-3y^2\right)}{44} - \frac{7\sqrt{55}G_{3s}yz\left(3x^2-y^2\right)}{44} - \frac{5\sqrt{33}G_{3x}xz\left(3x^2+3y^2-4z^2\right)}{132} - \frac{5\sqrt{33}G_{3y}yz\left(3x^2+3y^2-4z^2\right)}{132}$$
$$+ \frac{\sqrt{330}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{132} + \frac{\sqrt{22}G_z^{\alpha}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{88} + \frac{\sqrt{330}G_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{7\sqrt{4290}G_{3c}xz\left(x^2-3y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288}-\frac{7\sqrt{4290}G_{3s}yz\left(3x^2-y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288}$$
$$+\frac{35\sqrt{286}G_{3x}xz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}$$
$$+\frac{35\sqrt{286}G_{3y}yz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}+\frac{7\sqrt{715}G_{xyz}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}$$
$$-\frac{5\sqrt{429}G_z^{\alpha}\left(5x^6+15x^4y^2-90x^4z^2+15x^2y^4-180x^2y^2z^2+120x^2z^4+5y^6-90y^4z^2+120y^2z^4-16z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_{1g})$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$G_{3s}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_{1g})$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{15}G_{3s}\left(x^2+y^2-2z^2\right)}{12}-\frac{G_{3x}xy}{2}-\frac{G_{3y}\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}G_{xyz}xz}{4}+\frac{\sqrt{10}G_z^{\beta}yz}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_{1g})$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{22}G_{3s}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{176}-\frac{\sqrt{330}G_{3x}xy\left(5x^2-2y^2-9z^2\right)}{66}+\frac{\sqrt{330}G_{3y}\left(x^4-42x^2y^2+36x^2z^2+13y^4-36y^2z^2\right)}{528}$$
$$+\frac{5\sqrt{33}G_{xyz}xz\left(3x^2+3y^2-4z^2\right)}{132}-\frac{7\sqrt{55}G_z^{\alpha}yz\left(3x^2-y^2\right)}{44}+\frac{5\sqrt{33}G_z^{\beta}yz\left(3x^2+3y^2-4z^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_{1g})$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{7\sqrt{429}G_{3c}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{208}-\frac{\sqrt{429}G_{3s}\left(113x^6-1740x^4y^2+45x^4z^2+1725x^2y^4+90x^2y^2z^2-60x^2z^4-118y^6+45y^4z^2-60y^2z^4+8z^6\right)}{6864}$$
$$-\frac{7\sqrt{715}G_{3x}xy\left(7x^4+2x^2y^2-76x^2z^2-5y^4+44y^2z^2+16z^4\right)}{2288}$$
$$+\frac{7\sqrt{715}G_{3y}\left(x^6-8x^4y^2-7x^4z^2-7x^2y^4+90x^2y^2z^2-8x^2z^4+2y^6-23y^4z^2+8y^2z^4\right)}{2288}$$
$$-\frac{7\sqrt{286}G_{xyz}xz\left(7x^4-85x^2y^2+5x^2z^2+40y^4+5y^2z^2-2z^4\right)}{1144}-\frac{7\sqrt{4290}G_z^{\alpha}yz\left(3x^2-y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288}$$
$$+\frac{7\sqrt{286}G_z^{\beta}yz\left(85x^4-160x^2y^2-10x^2z^2+19y^4-10y^2z^2+4z^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_{2g})$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$G_{3c}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_{2g})$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{15}G_{3c}\left(x^2+y^2-2z^2\right)}{12} - \frac{G_{3x}\left(x-y\right)\left(x+y\right)}{4} + \frac{G_{3y}xy}{2} - \frac{\sqrt{10}G_{xyz}yz}{4} + \frac{\sqrt{10}G_z^\beta xz}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_{2g})$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{22}G_{3c}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{176} - \frac{\sqrt{330}G_{3x}\left(13x^4-42x^2y^2-36x^2z^2+y^4+36y^2z^2\right)}{528}$$
$$- \frac{\sqrt{330}G_{3y}xy\left(2x^2-5y^2+9z^2\right)}{66} - \frac{5\sqrt{33}G_{xyz}yz\left(3x^2+3y^2-4z^2\right)}{132} - \frac{7\sqrt{55}G_z^\alpha xz\left(x^2-3y^2\right)}{44} + \frac{5\sqrt{33}G_z^\beta xz\left(3x^2+3y^2-4z^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_{2g})$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{429}G_{3c}\left(118x^6-1725x^4y^2-45x^4z^2+1740x^2y^4-90x^2y^2z^2+60x^2z^4-113y^6-45y^4z^2+60y^2z^4-8z^6\right)}{6864}$$
$$+ \frac{7\sqrt{429}G_{3s}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{208} - \frac{7\sqrt{715}G_{3x}\left(2x^6-7x^4y^2-23x^4z^2-8x^2y^4+90x^2y^2z^2+8x^2z^4+y^6-7y^4z^2-8y^2z^4\right)}{2288}$$
$$- \frac{7\sqrt{715}G_{3y}xy\left(5x^4-2x^2y^2-44x^2z^2-7y^4+76y^2z^2-16z^4\right)}{2288} + \frac{7\sqrt{286}G_{xyz}yz\left(40x^4-85x^2y^2+5x^2z^2+7y^4+5y^2z^2-2z^4\right)}{1144}$$
$$- \frac{7\sqrt{4290}G_z^\alpha xz\left(x^2-3y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288} + \frac{7\sqrt{286}G_z^\beta xz\left(19x^4-160x^2y^2-10x^2z^2+85y^4-10y^2z^2+4z^4\right)}{2288}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-3)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(3,-3)}[g](E_{1g})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$G_{3x}$$

$$G_{3y}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-1)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(3,-1)}[g](E_{1g})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$-\frac{G_{3c}\left(x-y\right)\left(x+y\right)}{4} - \frac{G_{3s}xy}{2} + \frac{\sqrt{15}G_{3x}\left(x^2-3y^2+2z^2\right)}{20} + \frac{\sqrt{15}G_{3y}xy}{5} + \frac{\sqrt{6}G_{xyz}yz}{4} + \frac{\sqrt{10}G_z^\alpha xz}{10} + \frac{\sqrt{6}G_z^\beta xz}{4}$$

$$\frac{G_{3c}xy}{2} - \frac{G_{3s}\left(x-y\right)\left(x+y\right)}{4} + \frac{\sqrt{15}G_{3x}xy}{5} - \frac{\sqrt{15}G_{3y}\left(3x^2-y^2-2z^2\right)}{20} + \frac{\sqrt{6}G_{xyz}xz}{4} + \frac{\sqrt{10}G_z^\alpha yz}{10} - \frac{\sqrt{6}G_z^\beta yz}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(3,1)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(3,1)}[g](E_{1g})$

** symmetry

$$-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$$

** expression

$$-\frac{\sqrt{330}G_{3c}\left(13x^4-42x^2y^2-36x^2z^2+y^4+36y^2z^2\right)}{528}-\frac{\sqrt{330}G_{3s}xy\left(5x^2-2y^2-9z^2\right)}{66}-\frac{\sqrt{22}G_{3x}\left(17x^4-6x^2y^2-96x^2z^2-23y^4+144y^2z^2-8z^4\right)}{528}$$
$$-\frac{5\sqrt{22}G_{3y}xy\left(x^2+y^2-6z^2\right)}{66}+\frac{\sqrt{55}G_{xyz}yz\left(9x^2-19y^2+16z^2\right)}{132}-\frac{5\sqrt{33}G_z^{\alpha}xz\left(3x^2+3y^2-4z^2\right)}{132}-\frac{\sqrt{55}G_z^{\beta}xz\left(5x^2+33y^2-16z^2\right)}{132}$$

$$-\frac{\sqrt{330}G_{3c}xy\left(2x^2-5y^2+9z^2\right)}{66}+\frac{\sqrt{330}G_{3s}\left(x^4-42x^2y^2+36x^2z^2+13y^4-36y^2z^2\right)}{528}$$
$$-\frac{5\sqrt{22}G_{3x}xy\left(x^2+y^2-6z^2\right)}{66}+\frac{\sqrt{22}G_{3y}\left(23x^4+6x^2y^2-144x^2z^2-17y^4+96y^2z^2+8z^4\right)}{528}$$
$$-\frac{\sqrt{55}G_{xyz}xz\left(19x^2-9y^2-16z^2\right)}{132}-\frac{5\sqrt{33}G_z^{\alpha}yz\left(3x^2+3y^2-4z^2\right)}{132}+\frac{\sqrt{55}G_z^{\beta}yz\left(33x^2+5y^2-16z^2\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(3,3)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(3,3)}[g](E_{1g})$

** symmetry

$$-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$$

** expression

$$-\frac{7\sqrt{715}G_{3c}\left(2x^6-7x^4y^2-23x^4z^2-8x^2y^4+90x^2y^2z^2+8x^2z^4+y^6-7y^4z^2-8y^2z^4\right)}{2288}$$
$$-\frac{7\sqrt{715}G_{3s}xy\left(7x^4+2x^2y^2-76x^2z^2-5y^4+44y^2z^2+16z^4\right)}{2288}$$
$$+\frac{5\sqrt{429}G_{3x}\left(6x^6+11x^4y^2-101x^4z^2+4x^2y^4-90x^2y^2z^2+116x^2z^4-y^6+11y^4z^2+4y^2z^4-8z^6\right)}{2288}$$
$$+\frac{35\sqrt{429}G_{3y}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{2288}-\frac{7\sqrt{4290}G_{xyz}yz\left(8x^4+7x^2y^2-23x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}$$
$$+\frac{35\sqrt{286}G_z^{\alpha}xz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}-\frac{7\sqrt{4290}G_z^{\beta}xz\left(7x^4-4x^2y^2-22x^2z^2-11y^4+26y^2z^2+4z^4\right)}{2288}$$

$$-\frac{7\sqrt{715}G_{3c}xy\left(5x^4-2x^2y^2-44x^2z^2-7y^4+76y^2z^2-16z^4\right)}{2288}$$
$$+\frac{7\sqrt{715}G_{3s}\left(x^6-8x^4y^2-7x^4z^2-7x^2y^4+90x^2y^2z^2-8x^2z^4+2y^6-23y^4z^2+8y^2z^4\right)}{2288}$$
$$+\frac{35\sqrt{429}G_{3x}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{2288}$$
$$-\frac{5\sqrt{429}G_{3y}\left(x^6-4x^4y^2-11x^4z^2-11x^2y^4+90x^2y^2z^2-4x^2z^4-6y^6+101y^4z^2-116y^2z^4+8z^6\right)}{2288}$$
$$+\frac{7\sqrt{4290}G_{xyz}xz\left(x^4-7x^2y^2-x^2z^2-8y^4+23y^2z^2-2z^4\right)}{1144}+\frac{35\sqrt{286}G_z^{\alpha}yz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}$$
$$-\frac{7\sqrt{4290}G_z^{\beta}yz\left(11x^4+4x^2y^2-26x^2z^2-7y^4+22y^2z^2-4z^4\right)}{2288}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-3)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(3,-3)}[g](E_{2g})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$G_{xyz}$$

$$G_z^{\beta}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-1)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(3,-1)}[g](E_{2g})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{10}G_{3c}yz}{4} + \frac{\sqrt{10}G_{3s}xz}{4} + \frac{\sqrt{6}G_{3x}yz}{4} + \frac{\sqrt{6}G_{3y}xz}{4} - G_z^\alpha xy$$

$$\frac{\sqrt{10}G_{3c}xz}{4} + \frac{\sqrt{10}G_{3s}yz}{4} + \frac{\sqrt{6}G_{3x}xz}{4} - \frac{\sqrt{6}G_{3y}yz}{4} - \frac{G_z^\alpha (x-y)(x+y)}{2}$$

$\vec{\mathbb{G}}_{3,1}^{(3,1)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(3,1)}[g](E_{2g})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{5\sqrt{33}G_{3c}yz\left(3x^2+3y^2-4z^2\right)}{132} + \frac{5\sqrt{33}G_{3s}xz\left(3x^2+3y^2-4z^2\right)}{132} + \frac{\sqrt{55}G_{3x}yz\left(9x^2-19y^2+16z^2\right)}{132} - \frac{\sqrt{55}G_{3y}xz\left(19x^2-9y^2-16z^2\right)}{132}$$
$$- \frac{7\sqrt{22}G_{xyz}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{66} + \frac{\sqrt{330}G_z^\alpha xy\left(x^2+y^2-6z^2\right)}{132} + \frac{35\sqrt{22}G_z^\beta xy(x-y)(x+y)}{132}$$

$$\frac{5\sqrt{33}G_{3c}xz\left(3x^2+3y^2-4z^2\right)}{132} + \frac{5\sqrt{33}G_{3s}yz\left(3x^2+3y^2-4z^2\right)}{132} - \frac{\sqrt{55}G_{3x}xz\left(5x^2+33y^2-16z^2\right)}{132} + \frac{\sqrt{55}G_{3y}yz\left(33x^2+5y^2-16z^2\right)}{132}$$
$$+ \frac{35\sqrt{22}G_{xyz}xy(x-y)(x+y)}{132} + \frac{\sqrt{330}G_z^\alpha (x-y)(x+y)\left(x^2+y^2-6z^2\right)}{264} + \frac{7\sqrt{22}G_z^\beta\left(x^4-18x^2y^2+12x^2z^2+y^4+12y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{G}}_{3,1}^{(3,3)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(3,3)}[g](E_{2g})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{7\sqrt{286}G_{3c}yz\left(40x^4-85x^2y^2+5x^2z^2+7y^4+5y^2z^2-2z^4\right)}{1144} - \frac{7\sqrt{286}G_{3s}xz\left(7x^4-85x^2y^2+5x^2z^2+40y^4+5y^2z^2-2z^4\right)}{1144}$$
$$- \frac{7\sqrt{4290}G_{3x}yz\left(8x^4+7x^2y^2-23x^2z^2-y^4+y^2z^2+2z^4\right)}{1144} + \frac{7\sqrt{4290}G_{3y}xz\left(x^4-7x^2y^2-x^2z^2-8y^4+23y^2z^2-2z^4\right)}{1144}$$
$$+ \frac{\sqrt{429}G_{xyz}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{286}$$
$$+ \frac{7\sqrt{715}G_z^\alpha xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572} - \frac{21\sqrt{429}G_z^\beta xy(x-y)(x+y)\left(x^2+y^2-10z^2\right)}{572}$$

$$\frac{7\sqrt{286}G_{3c}xz\left(19x^4-160x^2y^2-10x^2z^2+85y^4-10y^2z^2+4z^4\right)}{2288} + \frac{7\sqrt{286}G_{3s}yz\left(85x^4-160x^2y^2-10x^2z^2+19y^4-10y^2z^2+4z^4\right)}{2288}$$
$$- \frac{7\sqrt{4290}G_{3x}xz\left(7x^4-4x^2y^2-22x^2z^2-11y^4+26y^2z^2+4z^4\right)}{2288} - \frac{7\sqrt{4290}G_{3y}yz\left(11x^4+4x^2y^2-26x^2z^2-7y^4+22y^2z^2-4z^4\right)}{2288}$$
$$- \frac{21\sqrt{429}G_{xyz}xy(x-y)(x+y)\left(x^2+y^2-10z^2\right)}{572} + \frac{7\sqrt{715}G_z^\alpha (x-y)(x+y)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$
$$- \frac{\sqrt{429}G_z^\beta\left(13x^6-45x^4y^2-150x^4z^2-45x^2y^4+540x^2y^2z^2+60x^2z^4+13y^6-150y^4z^2+60y^2z^4-8z^6\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](A_{2u})$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{2}G_{3x}y}{2} - \frac{\sqrt{2}G_{3y}x}{2}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](A_{2u})$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_{3c}y\left(3x^2-y^2\right)}{12}+\frac{\sqrt{15}G_{3s}x\left(x^2-3y^2\right)}{12}-\frac{G_{3x}y\left(x^2+y^2-4z^2\right)}{4}+\frac{G_{3y}x\left(x^2+y^2-4z^2\right)}{4}-\frac{\sqrt{10}G_{xyz}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}G_z^{\beta}xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{7\sqrt{330}G_{3c}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{528}+\frac{7\sqrt{330}G_{3s}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{528}+\frac{5\sqrt{22}G_{3x}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}$$
$$-\frac{5\sqrt{22}G_{3y}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}+\frac{7\sqrt{55}G_{xyz}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}-\frac{7\sqrt{55}G_z^{\beta}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_{1u})$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{3}G_{3c}z}{2}-\frac{\sqrt{2}G_{xyz}y}{4}+\frac{\sqrt{2}G_z^{\beta}x}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_{1u})$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{6}G_{3c}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}G_{3x}z\left(x-y\right)\left(x+y\right)}{4}-\frac{\sqrt{10}G_{3y}xyz}{2}-\frac{G_{xyz}y\left(x^2+y^2-4z^2\right)}{4}-\frac{\sqrt{15}G_z^{\alpha}x\left(x^2-3y^2\right)}{12}+\frac{G_z^{\beta}x\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_{1u})$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{33}G_{3c}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{528}+\frac{7\sqrt{55}G_{3x}z\left(5x^4-18x^2y^2-4x^2z^2+y^4+4y^2z^2\right)}{176}$$
$$+\frac{7\sqrt{55}G_{3y}xyz\left(x^2-2y^2+z^2\right)}{22}+\frac{\sqrt{22}G_{xyz}y\left(25x^4-55x^2y^2+15x^2z^2+4y^4+15y^2z^2-10z^4\right)}{88}$$
$$-\frac{7\sqrt{330}G_z^{\alpha}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{528}+\frac{\sqrt{22}G_z^{\beta}x\left(13x^4-100x^2y^2-30x^2z^2+55y^4-30y^2z^2+20z^4\right)}{176}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{3}G_{3s}z}{2}-\frac{\sqrt{2}G_{xyz}x}{4}-\frac{\sqrt{2}G_z^{\beta}y}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{6}G_{3s}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{10}G_{3x}xyz}{2}-\frac{\sqrt{10}G_{3y}z\left(x-y\right)\left(x+y\right)}{4}-\frac{G_{xyz}x\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{15}G_z^{\alpha}y\left(3x^2-y^2\right)}{12}-\frac{G_z^{\beta}y\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_{2u})$

** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{33}G_{3s}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{528} - \frac{7\sqrt{55}G_{3x}xyz\left(2x^2-y^2-z^2\right)}{22}$$
$$+\frac{7\sqrt{55}G_{3y}z\left(x^4-18x^2y^2+4x^2z^2+5y^4-4y^2z^2\right)}{176} + \frac{\sqrt{22}G_{xyz}x\left(4x^4-55x^2y^2+15x^2z^2+25y^4+15y^2z^2-10z^4\right)}{88}$$
$$+\frac{7\sqrt{330}G_z^{\alpha}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{528} - \frac{\sqrt{22}G_z^{\beta}y\left(55x^4-100x^2y^2-30x^2z^2+13y^4-30y^2z^2+20z^4\right)}{176}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,-2)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{3,2}^{(3,-2)}[g](E_{1u})$

** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$\frac{\sqrt{3}G_{3y}z}{6} - \frac{\sqrt{30}G_{xyz}x}{12} - \frac{\sqrt{2}G_z^{\alpha}y}{2} + \frac{\sqrt{30}G_z^{\beta}y}{12}$$

$$-\frac{\sqrt{3}G_{3x}z}{6} + \frac{\sqrt{30}G_{xyz}y}{12} + \frac{\sqrt{2}G_z^{\alpha}x}{2} + \frac{\sqrt{30}G_z^{\beta}x}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,0)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{3,2}^{(3,0)}[g](E_{1u})$

** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$\frac{\sqrt{10}G_{3c}xyz}{2} - \frac{\sqrt{10}G_{3s}z\left(x-y\right)\left(x+y\right)}{4} - \frac{\sqrt{6}G_{3y}z\left(3x^2+3y^2-2z^2\right)}{12} - \frac{\sqrt{15}G_{xyz}x\left(x^2-3y^2\right)}{12} + \frac{G_z^{\alpha}y\left(x^2+y^2-4z^2\right)}{4} + \frac{\sqrt{15}G_z^{\beta}y\left(3x^2-y^2\right)}{12}$$

$$\frac{\sqrt{10}G_{3c}z\left(x-y\right)\left(x+y\right)}{4} + \frac{\sqrt{10}G_{3s}xyz}{2} + \frac{\sqrt{6}G_{3x}z\left(3x^2+3y^2-2z^2\right)}{12} - \frac{\sqrt{15}G_{xyz}y\left(3x^2-y^2\right)}{12} - \frac{G_z^{\alpha}x\left(x^2+y^2-4z^2\right)}{4} - \frac{\sqrt{15}G_z^{\beta}x\left(x^2-3y^2\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,2)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{3,2}^{(3,2)}[g](E_{1u})$

** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$\frac{7\sqrt{55}G_{3c}xyz\left(2x^2-y^2-z^2\right)}{22} - \frac{7\sqrt{55}G_{3s}z\left(5x^4-18x^2y^2-4x^2z^2+y^4+4y^2z^2\right)}{176}$$
$$+\frac{5\sqrt{33}G_{3y}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{528} + \frac{\sqrt{330}G_{xyz}x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{264}$$
$$-\frac{5\sqrt{22}G_z^{\alpha}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176} - \frac{\sqrt{330}G_z^{\beta}y\left(15x^4+16x^2y^2-138x^2z^2+y^4-26y^2z^2+36z^4\right)}{528}$$

$$-\frac{7\sqrt{55}G_{3c}z\left(x^4-18x^2y^2+4x^2z^2+5y^4-4y^2z^2\right)}{176} - \frac{7\sqrt{55}G_{3s}xyz\left(x^2-2y^2+z^2\right)}{22}$$
$$-\frac{5\sqrt{33}G_{3x}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{528} + \frac{\sqrt{330}G_{xyz}y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{264}$$
$$+\frac{5\sqrt{22}G_z^{\alpha}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176} - \frac{\sqrt{330}G_z^{\beta}x\left(x^4+16x^2y^2-26x^2z^2+15y^4-138y^2z^2+36z^4\right)}{528}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,-2)}[g](E_{2u})$, $\vec{\mathbb{Q}}_{3,2}^{(3,-2)}[g](E_{2u})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{2}G_{3c}x}{4} + \frac{\sqrt{2}G_{3s}y}{4} + \frac{\sqrt{30}G_{3x}x}{12} - \frac{\sqrt{30}G_{3y}y}{12} - \frac{\sqrt{3}G_z^\beta z}{3}$$

$$\frac{\sqrt{2}G_{3c}y}{4} - \frac{\sqrt{2}G_{3s}x}{4} - \frac{\sqrt{30}G_{3x}y}{12} - \frac{\sqrt{30}G_{3y}x}{12} + \frac{\sqrt{3}G_{xyz}z}{3}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,0)}[g](E_{2u})$, $\vec{\mathbb{Q}}_{3,2}^{(3,0)}[g](E_{2u})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{G_{3c}x\left(x^2+y^2-4z^2\right)}{4} + \frac{G_{3s}y\left(x^2+y^2-4z^2\right)}{4} + \frac{\sqrt{15}G_{3x}x\left(x^2-3y^2\right)}{12} + \frac{\sqrt{15}G_{3y}y\left(3x^2-y^2\right)}{12} + \frac{\sqrt{10}G_z^\alpha z\left(x-y\right)\left(x+y\right)}{4} + \frac{\sqrt{6}G_z^\beta z\left(3x^2+3y^2-2z^2\right)}{12}$$

$$\frac{G_{3c}y\left(x^2+y^2-4z^2\right)}{4} - \frac{G_{3s}x\left(x^2+y^2-4z^2\right)}{4} - \frac{\sqrt{15}G_{3x}y\left(3x^2-y^2\right)}{12} + \frac{\sqrt{15}G_{3y}x\left(x^2-3y^2\right)}{12} - \frac{\sqrt{6}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12} - \frac{\sqrt{10}G_z^\alpha xyz}{2}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,2)}[g](E_{2u})$, $\vec{\mathbb{Q}}_{3,2}^{(3,2)}[g](E_{2u})$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{22}G_{3c}x\left(4x^4-55x^2y^2+15x^2z^2+25y^4+15y^2z^2-10z^4\right)}{88} - \frac{\sqrt{22}G_{3s}y\left(25x^4-55x^2y^2+15x^2z^2+4y^4+15y^2z^2-10z^4\right)}{88}$$
$$-\frac{\sqrt{330}G_{3x}x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{264} - \frac{\sqrt{330}G_{3y}y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{264}$$
$$-\frac{7\sqrt{55}G_z^\alpha z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44} + \frac{\sqrt{33}G_z^\beta z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}$$

$$\frac{\sqrt{22}G_{3c}y\left(55x^4-100x^2y^2-30x^2z^2+13y^4-30y^2z^2+20z^4\right)}{176} - \frac{\sqrt{22}G_{3s}x\left(13x^4-100x^2y^2-30x^2z^2+55y^4-30y^2z^2+20z^4\right)}{176}$$
$$+\frac{\sqrt{330}G_{3x}y\left(15x^4+16x^2y^2-138x^2z^2+y^4-26y^2z^2+36z^4\right)}{528} + \frac{\sqrt{330}G_{3y}x\left(x^4+16x^2y^2-26x^2z^2+15y^4-138y^2z^2+36z^4\right)}{528}$$
$$-\frac{\sqrt{33}G_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132} + \frac{7\sqrt{55}G_z^\alpha xyz\left(x^2+y^2-2z^2\right)}{22}$$